

USER MANUAL

This guide covers the deployment, launch, and operation of a traffic analysis and network threat alert system based on machine learning algorithms.

This guide will cover an example of deploying the system using Docker Compose and Docker containers. A similar deployment can also be performed, for example, in a Kubernetes environment.

To run Docker Compose, you must install Docker and all required components according to the official documentation.

We recommend first deploying the required environment and, after verifying its correct operation and configuration, proceeding with the system deployment.

1.1 System requirements

The following recommended system requirements were defined for the traffic analysis and threat detection system based on machine learning algorithms:

- OS: Linux (Ubuntu 20.04 LTS and above), Windows 10 and above (limited), macOS Ventura and above (limited);
- CPU: AMD Ryzen 5600 or Intel Core i5-10400;
- RAM: 16 GB DDR4;
- Disk: 30 GB (preferably SSD);
- Browser: Chrome, Opera, Firefox, Safari;

1.2 Existing restrictions

Modules designed to intercept traffic and obtain information about the host system require the use of a Docker host network. However, this configuration is only supported on Linux-based operating systems due to the specifics of the Docker architecture.

Therefore, there are no limitations when deploying the system in a Linux environment. However, attempting to run these modules in containers running Windows or macOS will result in errors.

This limitation can be circumvented by running these modules as standalone applications outside of containers.

1.3 Deploying the environment

For the system to function properly, a number of services are required in the environment. Some services are essential and some are optional.

Required services:

- RabbitMQ – used for exchanging information between services;
- Redis – used for storing data.

Optional services:

- RedisInsight – used for viewing the contents of the Redis storage;
- Prometheus – used for collecting metrics;

– Grafana – used for visualizing metrics.

Without Prometheus or Grafana, the user interface will not be able to view graphs and usage statistics.

1.3.1 RabbitMQ

It's recommended to run the message broker as a container. This ensures a portable environment and is easily reproducible.

RabbitMQ service startup description:

```
rabbitmq:
  image: rabbitmq:management
  container_name: rabbitmq
  volumes:
    -
      ./rabbitmq/rabbit_conf.json:/etc/rabbitmq/definitions.json
      - ./rabbitmq/data:/var/lib/rabbitmq
    environment:
      RABBITMQ_SERVER_ADDITIONAL_ERL_ARGS: "-
rabbitmq_management load_definitions
\"/etc/rabbitmq/definitions.json\""
    ports:
      - "5672:5672"
      - "15672:15672"
```

It is recommended to use the latest version with the control panel as the image. This method allows you to monitor the broker load. To save broker activity information, you must create persistent storage and assign it in the volumes section.

To access the broker control panel, you must forward container port 15672 to the host system. Port 5672 is the primary port and is required for RabbitMQ operation.

The broker requires a configuration that matches the system logic. The required configuration is available in the rabbit_conf.json file.

1.3.2 Redis

Redis storage must be launched in two instances, as the traffic interception and packet processing services require different storage.

Description of launching Redis instances:

```
redis-sniffer-svc:
  image: redis:latest
  ports:
    - "6379:6379"
  restart: always

redis-packet-processor-svc:
  image: redis:latest
  ports:
    - "6380:6379"
  restart: always
```

Within Docker, in this case, Redis stores are accessible at `redis://default@redis-sniffer-svc:6379` and `redis://default@redis-packet-processor-svc:6379`.

1.3.3 Redis Insight

Redis Insight is used to view the contents of Redis data stores and is not a required system component. Port 5540 is forwarded for external access.

Launch instructions:

```
redisinsight:
  image: redis/redisinsight:latest
  restart: always
  ports:
    - "5540:5540"
```

1.3.4 Prometheus

Prometheus is designed to collect metrics from running services. It makes requests to the `/metrics` URI and retrieves available metrics.

Deployment Description:

```
prometheus:
  image: prom/prometheus
  container_name: prometheus
  ports:
    - "9090:9090"
  volumes:
    - prometheus_data:/prometheus
  - ./monitoring/prometheus.yml:/etc/prometheus/prometheus.yml
```

Port 9090 is forwarded for external access. To permanently store collected metrics, a mapping is declared between directories on the host system and within the container.

The `prometheus_data` entity is a Docker object of type volume. It is declared in the volumes section of the Docker Compose file:

```
prometheus_data:
  driver: local
  driver_opts:
    o: bind
    type: none
    device: ./monitoring/prometheus_data
```

The `prometheus.yml` file describes the configuration, namely the services from which metrics should be obtained.

1.3.5 Prometheus Configuration File

The Prometheus configuration file is named `prometheus.yml`.

In the configuration file, you can configure the metrics collection interval:

```
global:
  scrape_interval: 5s
```

It contains a list of targets for which metrics should be collected. The targets have the following structure:

```
- job_name: "sniffer_service"
  metrics_path: "/metrics"
  static_configs:
    - targets: [ "host.docker.internal:8000" ]
```

In this configuration, `job_name` is the name of the target, `metrics_path` is the URI where the metrics are available, and the `targets` list specifies the address where the target is available.

1.3.6 Grafana

Grafana is used to visualize metrics and provide graphs. It depends on Prometheus and works in conjunction with it.

Grafana Launch Description:

```
grafana:
  image: grafana/grafana
  container_name: grafana
  ports:
    - "3000:3000"
  volumes:
    - grafana_data:/var/lib/Grafana
    - ./monitoring/grafana.ini:/etc/grafana/grafana.ini
```

To store data persistently, a mapping is declared between directories on the host system and within the container.

The `grafana_data` entity is a Docker object of type `volume`. It is declared in the `volumes` section of the Docker Compose file:

```
grafana_data:
  driver: local
  driver_opts:
    o: bind
    type: none
    device: ./monitoring/grafana_data
```

Port 3000 is available for accessing Grafana.

To support statistics graphs in the interface, you need to make the following changes to the `grafana.ini` configuration file:

```
allow_embedding = true
```

```

; auth_anonymous
enabled = true
org_name = <<organization name>>
org_role = Viewer

```

This configuration will allow access to charts through third-party resources.

1.4 System deployment

Each system module is a separate container capable of running independently.

This enables the system to run in distributed environments, such as Kubernetes.

System modules are deployed from a Docker Compose file, which describes specific parameters, environment variables, and settings specific to each module.

1.4.1 Traffic interception module

The traffic interception module depends on its own Redis storage and RabbitMQ message queue, which is specified in the `depends_on` section.

```

sniffer-svc:
  build: ./sniffer_svc
  network_mode: host
  depends_on:
    - redis-sniffer-svc
    - rabbitmq
  environment:
    - REDIS_HOST=localhost
    - REDIS_PORT=6379
    - ENABLE_METRICS=true
    - RABBITMQ_HOST=localhost
    - RABBITMQ_PORT=5672
    - RABBITMQ_USER=guest
    - RABBITMQ_PASSWORD=guest
  volumes:
    - ./public_key.pem:/app/public_key.pem:ro

```

The `build` section specifies the location of the Dockerfile for building the container with this service.

You also need to specify environment variables for connecting to RabbitMQ: `hostname`, `port`, `username`, and `password`.

Connection parameters to Redis are also specified using the corresponding environment variables.

The `ENABLE_METRICS` variable controls whether the service will return its metrics. It accepts the values `true` or `false`.

Note that the `network_mode` section specifies the host type, which is only available on Linux-based operating systems. This is discussed in more detail in Section 6.2.

1.4.2 Host System Information Provision Module

The host system information module has an environment variable, `ENABLE_METRICS`, which enables metrics in the service.

The build section specifies the location of the Dockerfile for building the container with this service.

```
system-info-svc:
  build: ./system_info_svc
  network_mode: host
  environment:
    - ENABLE_METRICS=true
  volumes:
    - ./public_key.pem:/app/public_key.pem:ro
```

1.4.3 Network packet processing module

The network packet processing module depends on its own Redis and RabbitMQ storage. Connection parameters are stored in environment variables.

The build section specifies the directory containing the Dockerfile for the service.

```
packet-processor-svc:
  build: ./packet_processor_svc
  ports:
    - "8002:8002"
  depends_on:
    - redis-packet-processor-svc
    - rabbitmq
  environment:
    - REDIS_HOST=redis-packet-processor-svc
    - REDIS_PORT=6379
    - ENABLE_METRICS=true
    - RABBITMQ_HOST=rabbitmq
    - RABBITMQ_PORT=5672
    - RABBITMQ_USER=guest
    - RABBITMQ_PASSWORD=guest
  volumes:
    - ./public_key.pem:/app/public_key.pem:ro
```

1.4.4 Data processing module for machine learning model

The module depends on RabbitMQ, the connection parameters are specified in the corresponding environment variables.

```
ml-data-processor-svc:
  build: ./ml_data_processor_svc
  ports:
    - "8003:8003"
  depends_on:
    - rabbitmq
  environment:
    - ENABLE_METRICS=true
```

```

- RABBITMQ_HOST=rabbitmq
- RABBITMQ_PORT=5672
- RABBITMQ_USER=guest
- RABBITMQ_PASSWORD=guest
volumes:
- ./public_key.pem:/app/public_key.pem:ro

```

1.4.5 Machine Learning Model Training Module

The machine learning model training module depends on RabbitMQ. Connection parameters are specified in the corresponding environment variables.

```

ml-training-svc:
  build: ./ml_training_svc
  ports:
    - "8004:8004"
  depends_on:
    - rabbitmq
  environment:
    - ENABLE_METRICS=true
    - RABBITMQ_HOST=rabbitmq
    - RABBITMQ_PORT=5672
    - RABBITMQ_USER=guest
    - RABBITMQ_PASSWORD=guest
  volumes:
    - ./public_key.pem:/app/public_key.pem:ro

```

1.4.6 Threat Detection Module

The threat detection module also depends on RabbitMQ. The broker connection parameters are specified in environment variables.

```

threat-detection-svc:
  build: ./threat_detection_svc
  ports:
    - "8005:8005"
  depends_on:
    - rabbitmq
  environment:
    - ENABLE_METRICS=true
    - RABBITMQ_HOST=rabbitmq
    - RABBITMQ_PORT=5672
    - RABBITMQ_USER=guest
    - RABBITMQ_PASSWORD=guest
  volumes:
    - ./public_key.pem:/app/public_key.pem:ro

```

1.4.7 Threat Detection Notification Module

The module for alerting on detected threats is deployed similarly to the previous modules:

```

notification-svc:
  build: ./notification_svc

```

```

ports:
  - "8006:8006"
environment:
  - ENABLE_METRICS=true
volumes:
  - ./public_key.pem:/app/public_key.pem:ro
  - ./private_key.pem:/app/private_key.pem:ro

```

1.4.7 Authorization service

The authorization service is responsible for generating access tokens. To ensure token uniqueness, a key pair is used for encryption.

The files containing the keys are specified in the volumes section.

```

auth-svc:
  build: ./auth_svc
  ports:
    - "8008:8008"
  environment:
    - ENABLE_METRICS=true
  volumes:
    - ./public_key.pem:/app/public_key.pem:ro
    - ./private_key.pem:/app/private_key.pem:ro

```

1.4.8 Example of module assembly

Each module is built from its own Dockerfile. Each module has a Dockerfile that looks something like this, with the first section responsible for building and compiling files:

```

FROM python:3.12-alpine AS builder
WORKDIR /app
RUN apk add --no-cache gcc musl-dev linux-headers
COPY requirements.txt .
RUN pip install --user --no-cache-dir -r requirements.txt

```

The second part copies only the assembled components with the required dependencies, which allows to reduce the size of the final container.

```

FROM python:3.12-alpine
WORKDIR /app
COPY --from=builder /root/.local /root/.local
COPY api/ ./api/
ENV PATH=/root/.local/bin:$PATH
EXPOSE 8000
CMD ["python", "-m", "api.main"]

```

1.5 Authorization and authentication

There are three user roles:

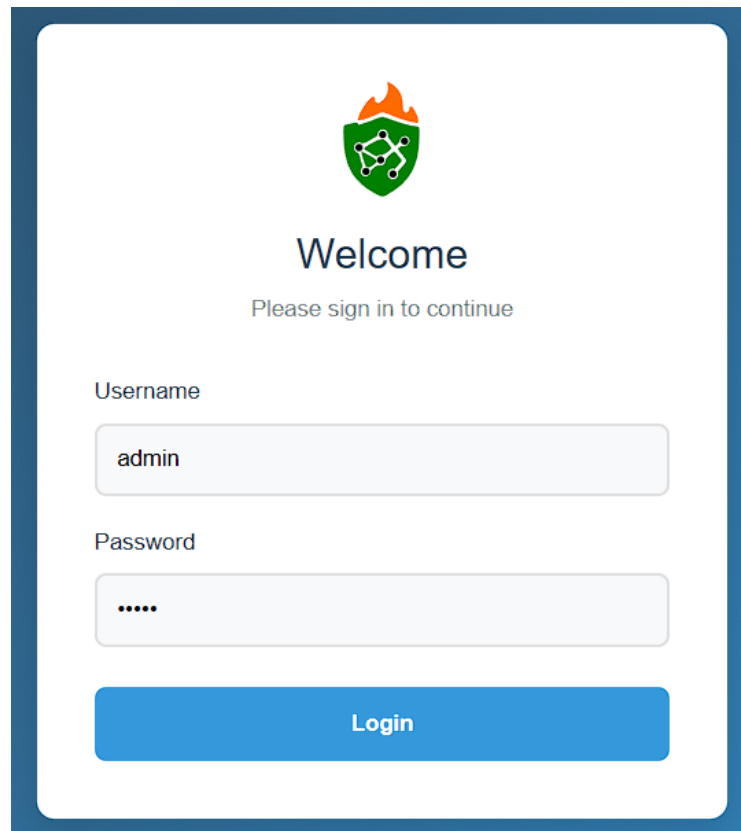
- Administrator;
- User;

– Viewer.

The roles differ in the available actions and capabilities.

To add users, log in with the username and password "admin" during the first launch. We recommend replacing the default administrator account with your own.

Figure 6.1 shows the authorization interface.



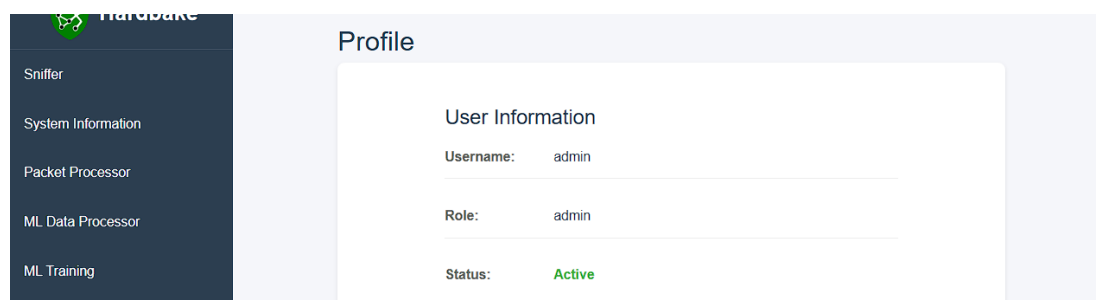
The image shows a login interface with a blue border. At the top center is a logo consisting of a green shield with a white molecular structure inside, and an orange flame above it. Below the logo, the word "Welcome" is displayed in a large, dark blue font. Underneath "Welcome" is the text "Please sign in to continue" in a smaller, gray font. There are two input fields: the first is labeled "Username" and contains the text "admin"; the second is labeled "Password" and contains six dots. Below these fields is a large blue button with the word "Login" in white text.

Figure 6.1 – Authorization window

To view detailed information about your current profile, go to the "Profile" tab in the navigation panel.

This will display information about your username, current role, and profile status.

The tab's appearance is shown in Figure 6.2.



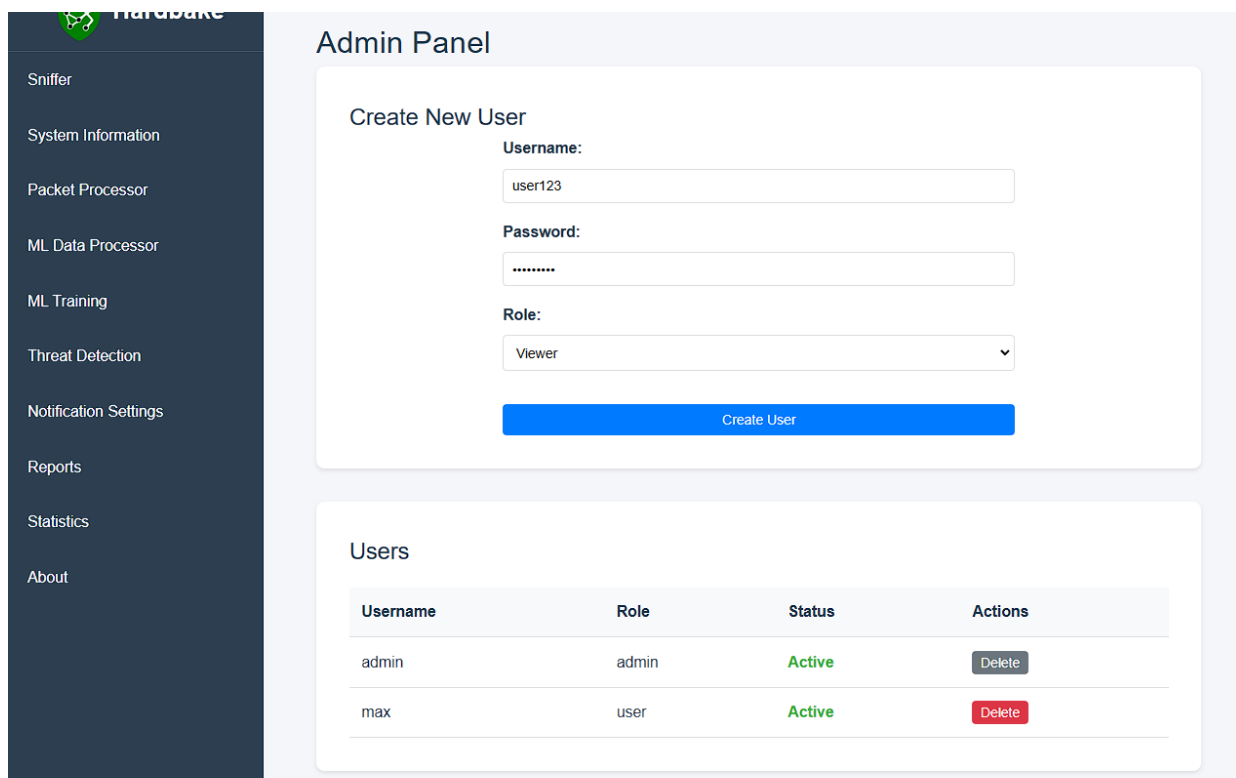
The image shows a web application interface. On the left is a dark blue navigation panel with a green logo at the top. The logo is a green shield with a white molecular structure and an orange flame. Below the logo, the word "Hardware" is written in white. Under "Hardware" are several menu items: "Sniffer", "System Information", "Packet Processor", "ML Data Processor", and "ML Training". The "Profile" tab is selected and highlighted in a light blue color. The main content area shows the "Profile" section. It has a title "Profile" in a dark blue font. Below the title is a white box with the heading "User Information". Under "User Information" are three rows of information: "Username: admin", "Role: admin", and "Status: Active". The "Status" is highlighted in green.

Figure 6.2 – Viewing profile information

If you're logged in with an administrator account, the "Admin Panel" tab will be available in the navigation bar.

In the Admin Panel, you can create new user accounts. To do so, enter the desired username and password, and select the desired role.

The Admin Panel is shown in Figure 6.3.



The screenshot displays the 'Admin Panel' interface. On the left is a dark sidebar with navigation links: Sniffer, System Information, Packet Processor, ML Data Processor, ML Training, Threat Detection, Notification Settings, Reports, Statistics, and About. The main content area is titled 'Admin Panel' and contains two sections. The top section, 'Create New User', has a form with fields for 'Username' (containing 'user123'), 'Password' (masked with dots), and 'Role' (a dropdown menu set to 'Viewer'). A blue 'Create User' button is at the bottom of this form. The bottom section, 'Users', contains a table with the following data:

Username	Role	Status	Actions
admin	admin	Active	<button>Delete</button>
max	user	Active	<button>Delete</button>

Figure 6.3 – Admin Panel

1.6 Starting traffic interception

Traffic interception is managed in the "Sniffer" tab. This tab is shown in Figure 6.4.

To start a traffic interception session, select the desired interface from the "Network Interface" drop-down list.

If necessary, you can specify the desired filtering parameters for incoming packets.

There are no default filter parameters. If no parameters are specified, it will be assumed that the user has not configured any.

Filtering by IP addresses, ports, and protocols is available.

If a parameter is specified that does not match the format, this will be indicated and the "Start Sniffing" button will be unclickable.

To save the interception result to a PCAP file, select the corresponding checkbox.

The "Refresh" button updates information about running traffic interception sessions.

To display sessions with a specific status, select the desired status from the

drop-down list.

To end a traffic interception session, find the desired session in the list and click the "Stop" button. The session state will change to "Stopped" and the session will end.

Hardbake

Sniffer

System Information

Packet Processor

ML Data Processor

ML Training

Threat Detection

Notification Settings

Reports

Statistics

About

Sniffer Service

Network Interface: Ethernet

☐ Write to File

Filter Parameters

Source IP: 192.168.1.1

Destination IP: 192.168.1.500
Invalid IPv4 address

Protocol: UDP

Source Port: 12345

Destination Port: 99999
Port must be between 0 and 65535

Start Sniffing Stop Sniffing

Sniffing Sessions

Filter by Status: All Statuses

Clear Refresh

Interface: Ethernet	Stop
Started: 09.05.2025, 18:13:01	
Status: Running	

Figure 6.4 – Menu for starting a traffic interception session

1.7 Viewing network interfaces and their parameters

The "System Information" tab displays information about the network interfaces available on the device and their parameters.

The system information tab is shown in Figure 6.5.

The "Refresh" button refreshes information about the host's network interfaces.

By default, detailed information is hidden. To expand the parameters for a selected interface, click the "Show Details" button.

To hide detailed information, click the "Hide Details" button.

The left side of the detailed information displays static interface parameters, such as IP addresses, network mask, and MAC address. If any of these parameters are missing, the missing parameters will not be displayed.

The right side displays dynamic interface parameters, such as the number of bytes and packets received/sent, errors, and dropped packets.

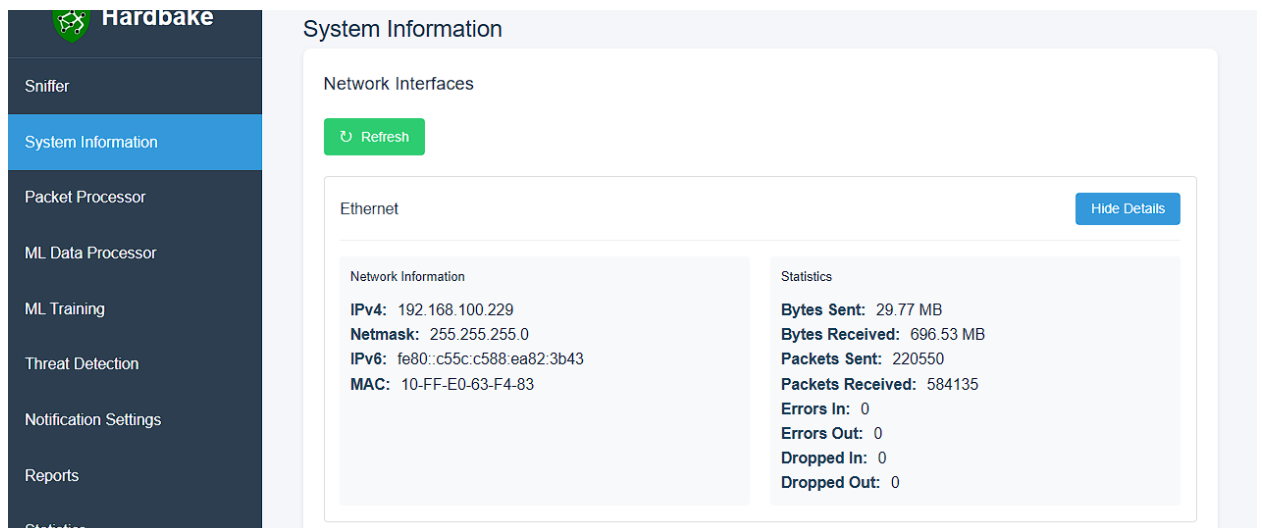


Figure 6.5 – Menu for viewing network interface parameters

1.8 Managing the Network Packet Processing Module

The "Packet Processor" tab controls packet processing. This tab is shown in Figure 6.6.

To upload a custom PCAP/PCAPNG traffic file, click the "Choose file..." button and select the desired file from disk. After selecting the file, click the "Upload" button.

Information about the uploaded file will appear in the "Recent Uploads" section with the "Running" status. After the status changes to "Processed," the number of TCP and UDP sessions detected in the uploaded file will appear in the information about the file.

Detected sessions can be sent for further processing using the "Send to RMQ" button. Data can only be sent for processing once. To view sessions with a specific processing status, select the desired status from the drop-down list.

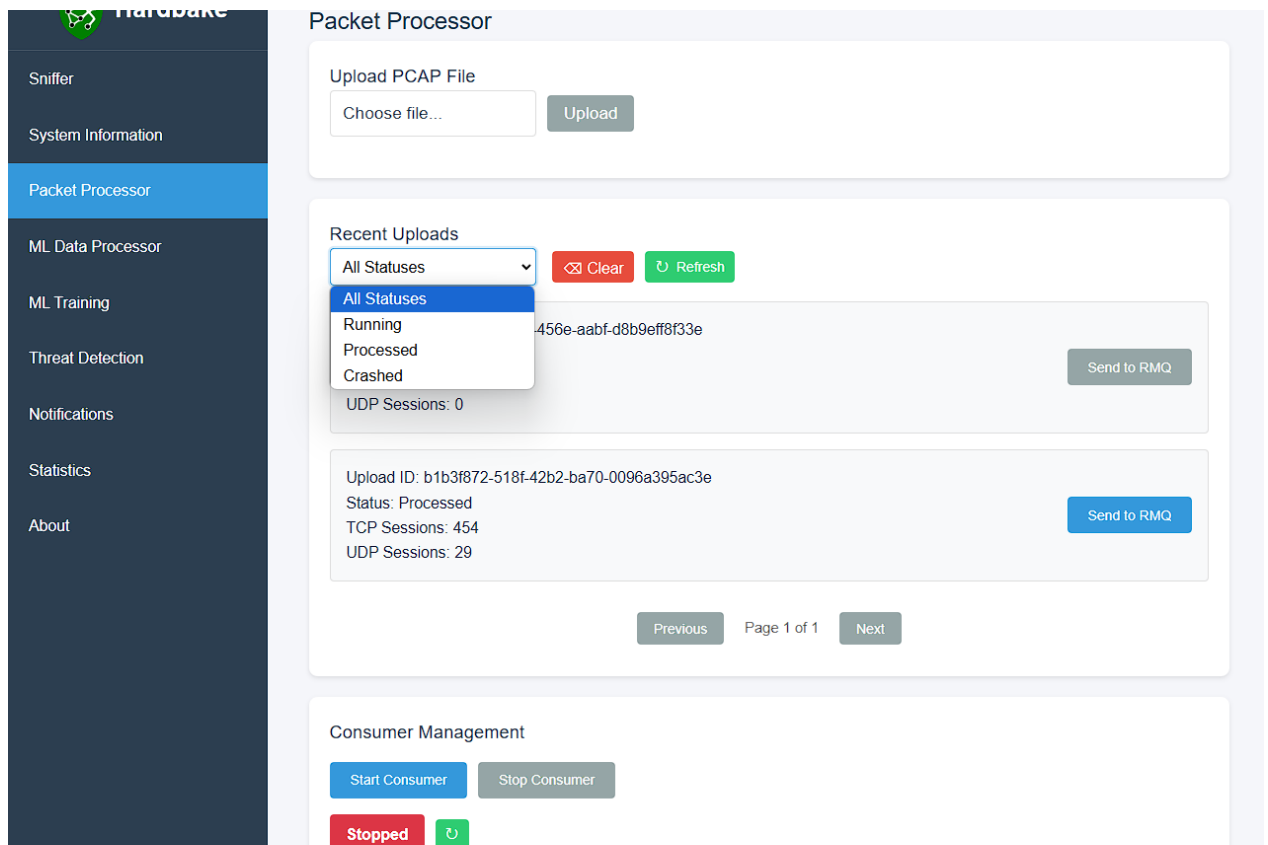


Figure 6.6 – Network packet processing module control menu

1.9 Managing the Data Processing Module for a Machine Learning Model

The "ML Data Processor" tab provides options for managing data processing for machine learning models. The tab's appearance is shown in Figure 6.7.

To start consuming batches, click the "Start Consumer" button. The status will change to "Running," and the data consumption process will begin. If errors occur during consumption, the status will change to "Error."

To stop batch processing, click the "Stop Consumer" button.

To update the consumer status, click the "Refresh" button, which will request the current consumer status.

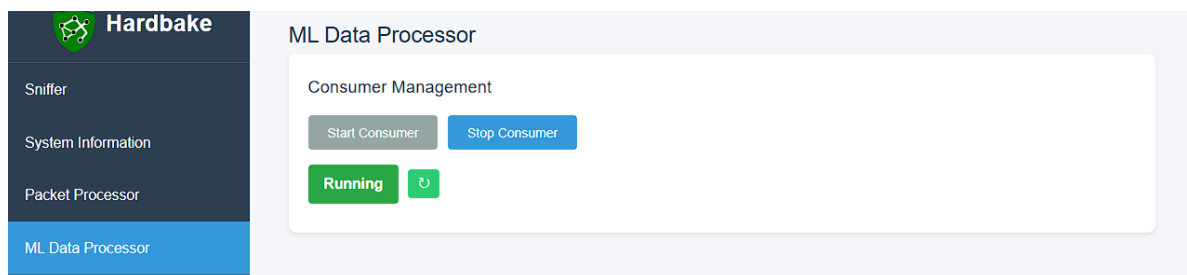


Figure 6.7 – Data processing module control menu for a machine learning model

1.10 Machine Learning Model Training Module

The "ML Training" tab is used to manage the training process of machine learning models and their parameters. This tab is shown in Figure 6.8.

The "Consumer Management" panel is used to manage data collection for model training.

To begin data collection, click the "Start Consuming" button. The status will change to Running, and data collection will begin. To stop data collection, click the "Stop Consuming" button.

The number of collected records for training is displayed in the "Collected Samples" parameter in the "Training Management" panel. To change the number of records for training, enter the desired number in the "Required Samples" field and click the "Update" button.

The "Start Training" button will be unavailable until the specified number of records has been collected.

The "Training Status" parameter displays the current training status of the model. If an error occurs, the status will change to "Error."

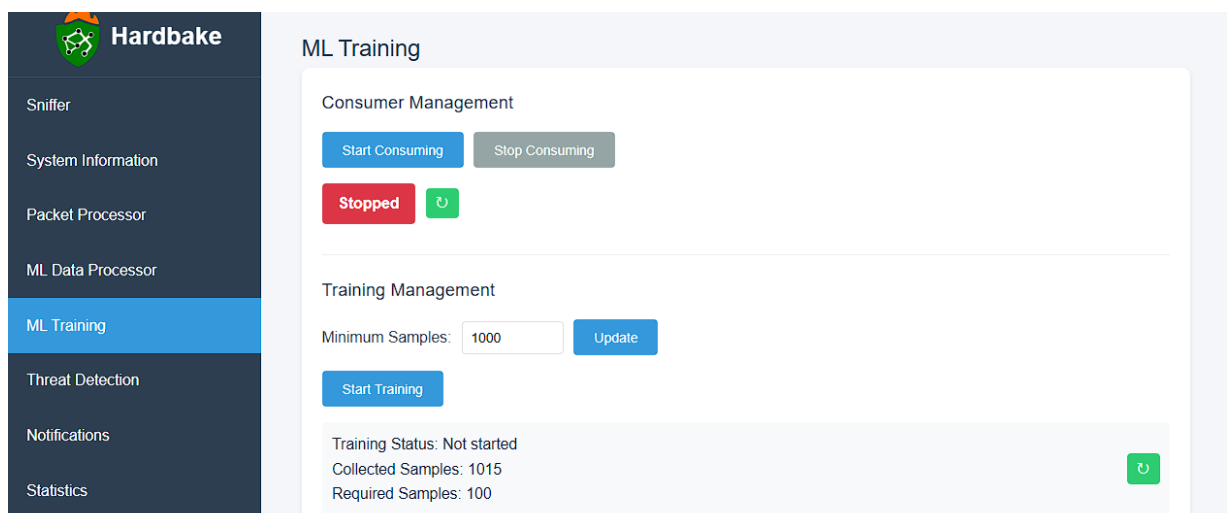


Figure 6.8 – Machine learning training module control menu

The "Model Settings" tab displays the current hyperparameters of machine learning models. The appearance of this tab is shown in Figure 6.9.

To refresh the displayed information about the current hyperparameters, click the "Refresh Settings" button.

To change the model hyperparameter values, click the "Update Parameters" button.

Models can be downloaded as files in the "Model Management" tab. The "Download Models" button becomes available if the model training status is "Ready."

The file will be downloaded to the directory specified by your browser as the downloaded files directory.

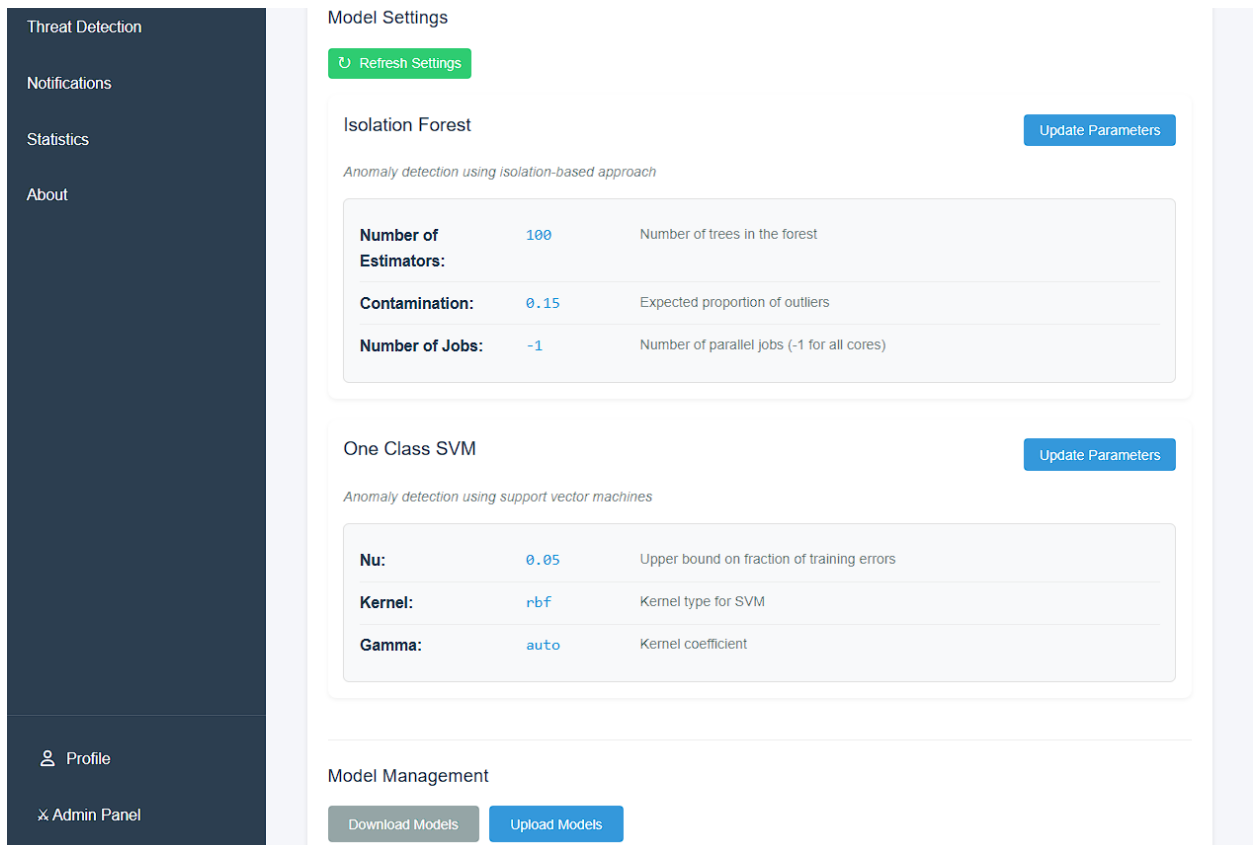


Figure 6.9 – Managing hyperparameters of machine learning models

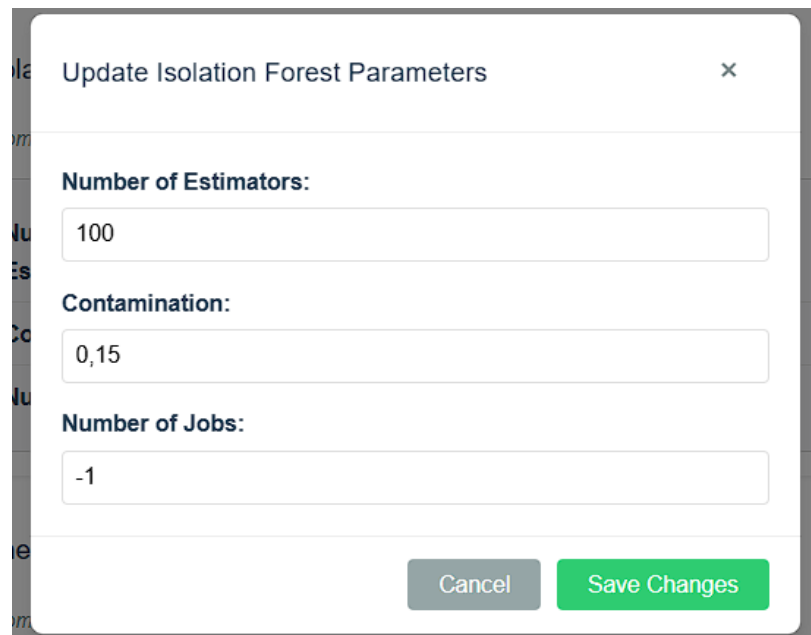
The hyperparameter update windows are shown in Figures 6.10 and 6.11.

The hyperparameters of the Isolation Forest model and their descriptions are listed below.

1. Number of Estimators. The number of trees in the ensemble. More trees result in higher accuracy and stability of results. Fewer trees result in faster training, but may result in a loss of quality.

2. Contamination. The proportion of anomalies in the sample (from 0 to 0.5). Determines the anomaly threshold: how many of the most isolated points are considered outliers. If "auto" is selected, the model will estimate the parameter automatically, but this will impact accuracy.

3. Number of Jobs. The number of cores used in parallel for training. A value of -1 uses all available cores.



Update Isolation Forest Parameters

Number of Estimators:

100

Contamination:

0,15

Number of Jobs:

-1

Cancel Save Changes

Figure 6.10 – Window for changing hyperparameters of the Isolation Forest model

The One Class SV model's hyperparameters and their descriptions are listed below.

1. Nu. Upper bound for the proportion of outliers and lower bound for the proportion of support vectors. Range: (0, 1]. Controls the model's sensitivity to outliers. For example, nu=0.1 means that no more than 10% of the data are outliers.

2. Kernel. Kernel function, which determines how the similarity of points is measured. It is recommended to use the "RBF" value, which works well with nonlinear data.

3. Gamma. It is recommended to start with "scale" and adjust it manually. The "auto" value will automatically select a value. A high value means the model is sensitive to local variations and may overfit. A low value means the model becomes "smooth" and may miss anomalies. This value should be positive.

Figure 6.11 – Window for changing hyperparameters of the One Class SVM model

1.11 Управление модулем обнаружения угроз

Вкладка «Threat Detection» предназначена для управления процессом анализа трафика на предмет угроз. Ее внешний вид показан на рисунке 6.12.

Для начала анализа нужно нажать кнопку «Start Detection». После этого, в случае успеха, статус измениться на «Running». В случае ошибки статус изменится на «Error». Для остановки процесса обнаружения угроз нужно нажать кнопку «Stop Detection».

Во вкладке «Batch Size Settings» можно изменить размер пакета данных для обработки. Для этого нужно ввести желаемый размер в поле «Batch Size» и нажать кнопку «Update».

Перед стартом процесса анализа, необходимо загрузить в модуль модель машинного обучения из файла, который можно скачать во вкладке «ML Training». При нажатии кнопки «Upload Models» будет предложено выбрать файл с диска.

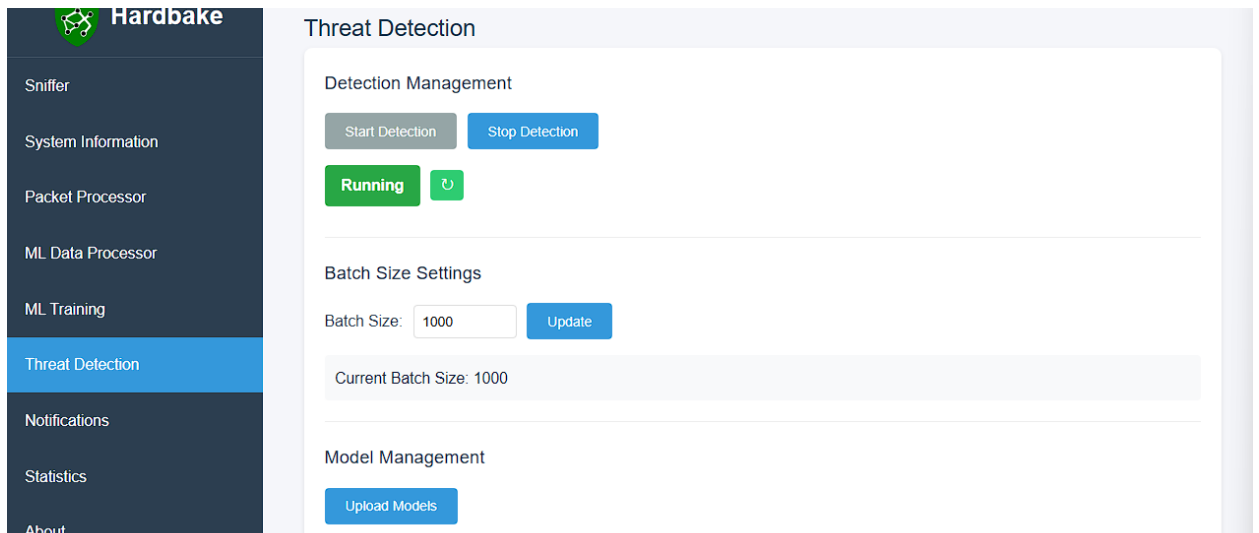


Figure 6.12 – Threat detection module control menu

1.12 Monitoring

The "Statistics" tab displays various system performance metrics in graph form.

The displayed parameters include statistics on requests to endpoints of various services.

Figure 6.13 shows an example of the displayed data.



Figure 6.13 – Statistics display page

1.13 Viewing detected threats

The "Notifications" tab displays information about detected suspicious connections.

To receive new data from the message queue, click the "Start Consuming" button. This will initiate the process of receiving information and writing it to the database. If an error occurs, the corresponding status will be displayed.

Filtering options are available for sorting and displaying the required

information. To apply filters, specify the desired values and click the "Apply Filters" button. To reset the filter fields, click the "Clear Filters" button.

The detected threats viewing tab is shown in Figure 6.14.

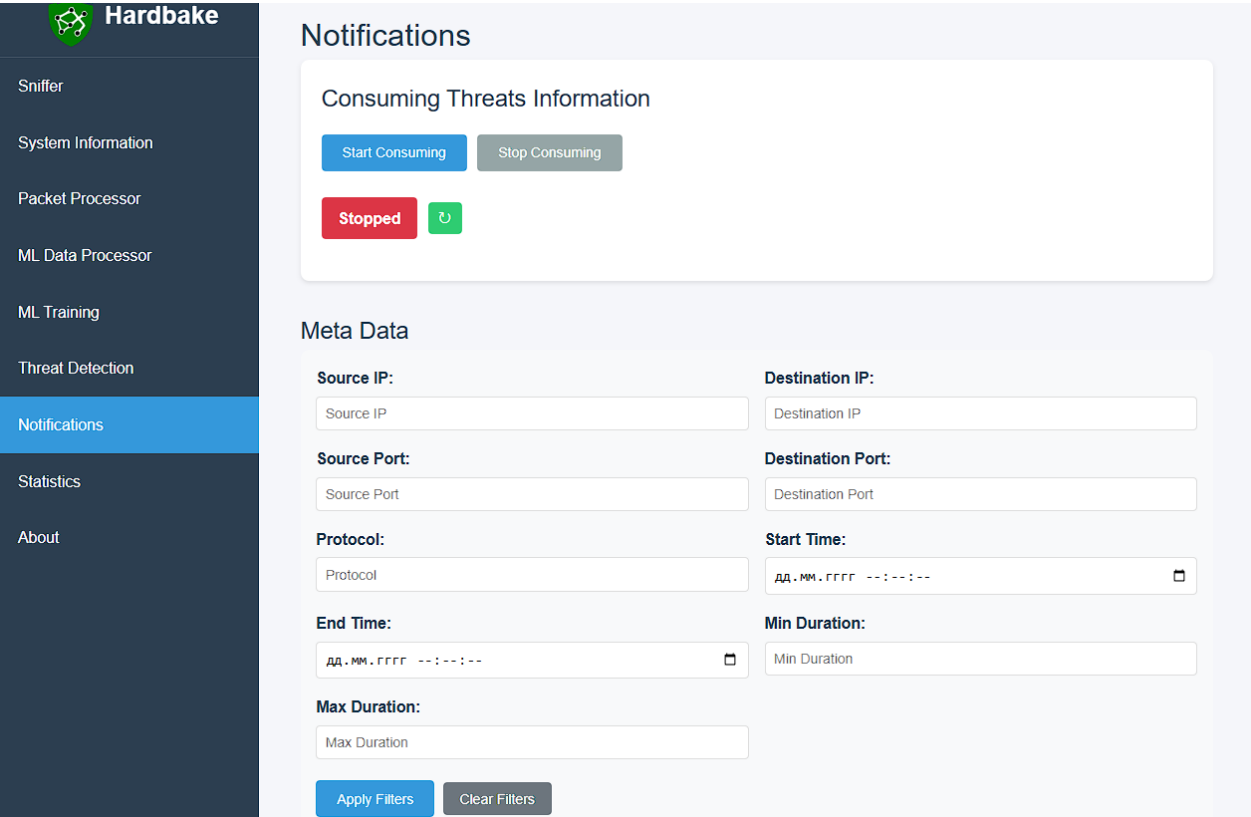


Figure 6.14 – Detected threats display page

Figure 6.15 shows an example of the displayed information about detected threats.

You can sort the information by column by clicking on the corresponding column heading, and an arrow will indicate the sort order.

ID	Source IP	Destination IP	Source Port	Destination Port	Protocol	Timestamp ▼	Duration	Actions
1063	192.168.100.229	52.72.201.59	65136	443	6	10.05.2025, 00:18:25	0.5387285	View Details
708	192.168.100.229	52.72.201.59	65135	443	6	10.05.2025, 00:18:25	0.5405602	View Details

Figure 6.15 – Displaying a list of detected threats

To view detailed session characteristics, click the "View Details" button. An example of a detailed output is shown in Figure 6.16.

Id:	708
Protocol:	6
Bwd Packet Length Max:	1466
Bwd Packet Length Min:	60
Bwd Packet Length Mean:	587.9000
Bwd Packet Length Std:	644.8644
Flow IAT Std:	54.9584
Flow IAT Max:	143
Fwd IAT Std:	69.4201
Fwd IAT Max:	144
Min Packet Length:	54
Max Packet Length:	1466
Packet Length Std:	560.1928
Packet Length Variance:	313815.9619
Psh Flag Count:	0
Avg Bwd Segment Size:	587.9000
Idle Min:	0
Idle Mean:	27.0280
Idle Max:	143

Figure 6.16 – Threat Details

Detailed characteristics include the parameters by which machine learning models classify compounds.

1.14 Changing the interface language

The system is localized into Russian and English. To switch languages, select the desired option from the drop-down list. Changing the language is shown in Figure 6.17.

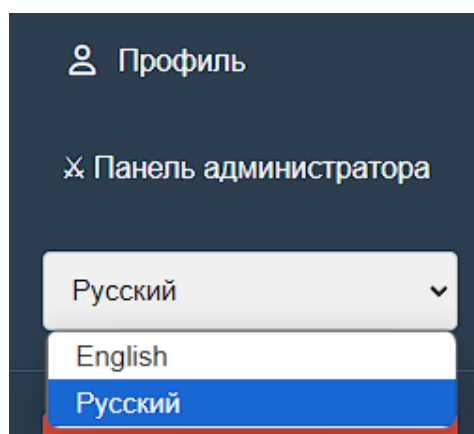


Figure 6.17 – Changing the interface language